

CONTACT INFORMATION	✉ E-mail: doserjef@msu.edu Tel: (585) 683-4170	Homepage: https://www.jeffdoser.com GitHub: https://github.com/doserjef
RESEARCH INTERESTS	Quantitative Ecology, Hierarchical Bayesian Modeling, Wildlife Ecology, Conservation Biology	
EDUCATION	Michigan State University, East Lansing, MI	
	Ph.D., Forestry and Ecology, Evolution, and Behavior	Completed 2022
	- Dissertation title: Development and application of hierarchical models for monitoring avian soundscapes, populations, and communities - Advisor: Andrew O. Finley	
	M.S., Applied Statistics	Completed 2021
	State University of New York at Geneseo, Geneseo, NY.	
	B.S. Mathematics and Biology Summa Cum Laude	Completed 2018
RESEARCH EXPERIENCE	Postdoctoral Research Associate, Michigan State University	2022 - Current
	- Department of Integrative Biology - Advisor: Elise F. Zipkin	
	Postdoctoral Fellow: Institute of Biodiversity, Ecology, Evolution, and Macrosystems, Michigan State University	2022 - Current
	Graduate Research Assistant, Michigan State University	2018-2022
	- Department of Forestry - Advisor: Andrew O. Finley	
	Undergraduate Research Assistant, SUNY Geneseo	2015-2018
	Advisors: Kristina M. Hannam, David D. Meisel	
TEACHING EXPERIENCE	Co-Instructor, Michigan State University	
	IBIO 831 Statistical Methods in Ecology and Evolution II	2022
	FOR/STT 875 R Programming for Data Sciences	2021-2022
	Teaching Assistant, Michigan State University	
	FOR/STT 875 R Programming for Data Sciences – Co-authored course textbook .	2018-2020
	FOR 472 Ecological Monitoring and Data Analysis	2018
	Undergraduate Lab Assistant, SUNY Geneseo	2016
	Introductory Biology Lab	
UNDER REVIEW	10. Doser, J.W. , Finley, A.O., Banerjee, S. Joint species distribution models with imperfect detection for high-dimensional spatial data. <i>Ecology</i> . Preprint: arXiv:2204.02707 .	
	9. Williamson, C.R., Campa III, H., Doser, J.W. , Winterstein S.R., Beyer Jr, D.E. Elk space-use and resource selection patterns in response to summer trail-based recreation. <i>Journal of Wildlife Management</i> .	

8. **Doser, J.W.**, Finley, A.O., Kéry, M., Zipkin, E.F. **spOccupancy**: An R package for single-species, multi-species, and integrated spatial occupancy models. *Methods in Ecology and Evolution*. <https://doi.org/10.1111/2041-210X.13897>.
7. **Doser, J.W.**, Leuenberger, W., Sillett, T.S., Hallworth, M.T., Zipkin, E.F. Integrated community occupancy models: A framework to assess occurrence and biodiversity dynamics using multiple data sources. *Methods in Ecology and Evolution*. 13(4), 919-932. <https://doi.org/10.1111/2041-210X.13811>.
6. **Doser, J.W.**, Weed, A.S., Zipkin, E.F., Miller, K.M., Finley, A.O. (2021). Trends in bird abundance differ among protected forests but not bird guilds. *Ecological Applications* 31(6):e02377. <https://esajournals.onlinelibrary.wiley.com/doi/abs/10.1002/eap.2377>.
5. **Doser, J. W.**, Finley, A. O., Weed, A. S., Zipkin, E. F. (2021). Integrating automated acoustic vocalization data and point count surveys for estimation of bird abundance. *Methods in Ecology and Evolution*, 12(6), 1040-1049. <https://doi.org/10.1111/2041-210X.13578>.
4. Hoffmann, H.M., Meadows, J.D., Breuer, J.A., Yaw, A.M., Nguyen, D., Tonsfeldt, K.J., Chin, A.Y., Devries, B.D., Khan, R., Trang, C., Oosterhouse, H.J., Sora Lee, J., **Doser, J.W.**, Gorman, M. R., Welsh, D.K., Mellon, P. L. (2021). The transcription factors VAX1 and SIX3 are required for suprachiasmatic nucleus circadian output and fertility in female mice. *Journal of Neuroscience Research*. <https://doi.org/10.1002/jnr.24864>.
3. Teimouri, M., **Doser, J.W.**, Finley A.O. (2020) **ForestFit**: An R package for modeling plant size distributions. *Environmental Modelling & Software*, 131, 104668. <https://doi.org/10.1016/j.envsoft.2020.104668>.
2. **Doser, J.W.**, Finley, A.O., Kasten, E.P., Gage, S.H. (2020). Assessing soundscape disturbance through hierarchical models and acoustic indices: A case study on a shelterwood logged northern Michigan forest. *Ecological Indicators*, 113, 106244. <https://doi.org/10.1016/j.ecolind.2020.106244>.
1. **Doser, J.W.**, Hannam, K.M., Finley, A.O., (2020). Characterizing functional relationships between anthropogenic and biological sounds: A western New York state soundscape case study. *Landscape Ecology*, 35(3), 689-707. <https://doi.org/10.1007/s10980-020-00973-2>.

BOOKS UNDER
CONTRACT

Finley, A.O. and **Doser, J.W.**. *Introduction to Forestry Data Analysis with R*. Chapman & Hall CRC. Started 2020.

PUBLISHED
SOFTWARE

spOccupancy. **Doser, J.W.**, Finley, A.O. (Nov 2021 - current). Single-species, Multi-species, and Integrated Spatial Occupancy Models. Downloaded 2674 times as of May 19, 2022. <https://cran.r-project.org/web/packages/spOccupancy/index.html>.

PROFESSIONAL
POSITIONS

Statistical Consultant, Michigan State University 2019-2020

- College of Agriculture and Natural Resources Statistical Consulting Center
- Provided statistical and programming guidance to undergraduate students, graduate students, and faculty members.

Student Database Programmer, SUNY Geneseo 2016-2018

- Computing and Information Technologies Department
- Developed, modified, and tested Banner applications using SQL, Groovy, SQR, and the Argos Enterprise Reporting System
- Experience in managing and testing Oracle databases.

RELEVANT
SKILLS

Programming Languages/Statistical Software: R, C++, C, JAGS, NIMBLE

Familiar Languages: SQL, Groovy, Matlab, Mathematica, SAS

Software: Latex/Knitr, R Markdown/bookdown, R Shiny

GRANTS AND SCHOLARSHIPS	Barry M. Goldwater Scholarship	2017
	Dr. Sorrell E. Chesin '58 Endowed Research Award	Fall 2016
	Dr. Wendell and Barbara Rhodes Research Award	Summer 2016
	Geneseo Foundation Scholarship	2014–2018
AWARDS AND HONORS	SUNY Chancellor's Award for Student Excellence	2018
	Excellence in Mathematics, SUNY Geneseo	2018
	Dr. Sam Molnar Scholar Athlete Award, SUNYAC	2018
	James Fulton Award for Academic and Athletic Excellence, SUNYAC	2018
	Phi Beta Kappa Honors Society, SUNY Geneseo	2017
	SUNY Chancellor's Scholar-Athlete Award	2016-2018
	Edgar Fellows Honors Program	2014–2018
UNIVERSITY SERVICE	Ecology, Evolution, and Behavior Program Research Symposium Chair, Michigan State University,	2021
	Ecology, Evolution, and Behavior Program Graduate/Postdoc Awards Committee, Michigan State University,	2020-current
	Ecology, Evolution, and Behavior Program Research Symposium Committee, Michigan State University	2021
	Treasurer, Graduate Student Organization, Department of Forestry, Michigan State University	2019-2020
	Student Athlete Advisory Committee, SUNY Geneseo	2015-2018
EDITORIAL SERVICE	Peer Reviews: <i>Nature Ecology and Evolution</i> (1), <i>Methods in Ecology and Evolution</i> (2), <i>Ecological Monographs</i> (1), <i>Ecological Applications</i> (1), <i>Ecology</i> (1), <i>Landscape Ecology</i> (1), <i>Biological Invasions</i> (1), <i>Ecological Indicators</i> (2), <i>Ecology and Evolution</i> (1), <i>Journal of Ornithology</i> (1), <i>PLOS ONE</i> (1)	
PRESENTATIONS	Doser, J. W. , Finley, A.O., Banerjee, S. August 2022. Joint species distribution models with imperfect detection for high-dimensional spatial data. Joint Statistical Meetings. Washington, D. C. Invited.	
	Doser, J. W. , Leuenberger, W., Sillett, T.S., Hallworth, M.T., Zipkin, E.F. June 2022. Integrated community occupancy models: A framework to assess occurrence and biodiversity dynamics using multiple data sources. International Statistical Ecology Conference. Remote.	
	Doser, J. W. . June 2022. Bayesian occupancy modeling with acoustic data in spOccupancy . Cornell Acoustic Methods Working Group Series. Remote. Invited.	
	Doser, J.W. , Weed, A.S., Zipkin, E.F., Miller, K.M., Finley, A.O. July 2021. Trends in bird abundance differ among protected forests but not bird guilds. American Ornithological Society Meeting. Remote.	
	Doser, J.W. , Finley, A.O., Kasten, E. P., Gage, S. H. June 2021. Assessing soundscape disturbance through hierarchical models and acoustic indices: A case study on a shelterwood logged northern Michigan forest. Oral Presentation. International Ecoacoustics Congress. Remote.	
	Doser, J.W. , Weed, A.S., Zipkin, E.F., Miller, K.M., Finley, A.O. May 2021. Trends in bird abundance differ among protected forests but not bird guilds. Ecology, Evolution, and Behavior Program Symposium. Michigan State University. Remote. Awarded 2nd Place for Best Oral Presentation.	
	Doser, J.W. February 2021. Forest bird monitoring using acoustic recordings and hierarchical models. Hanover Seminar Series. Department of Forestry. Michigan State University.	

- Doser, J.W.**, Finley, A.O., Kasten, E. P., Gage, S. H. April 2020. Assessing soundscape disturbance through hierarchical models and acoustic indices: A case study on a shelterwood logged northern Michigan forest. Oral Presentation. Conference on Applied Statistics in Agriculture and Natural Resources. Gainesville, FL (Conference canceled).
- Doser, J.W.**, Finley, A.O., Kasten, E. P., Gage, S. H. February 2020. Assessing soundscape disturbance through hierarchical models and acoustic indices: A case study on a shelterwood logged northern Michigan forest. Oral Presentation. Michigan State University.
- Doser, J.W.**, Hannam, K.M., Finley, A.O. September 2019. Using hierarchical Bayesian models to analyze acoustic data. Oral presentation, XXVII International Bioacoustics Congress, Brighton, England.
- Doser, J.W.**, Hannam, K.M., Finley, A.O. May 2019. Characterizing functional relationships between biophony and technophony: a western New York soundscape case study. Oral presentation, Northeast Regional Environmental Acoustics Symposium. Providence, RI.
- Doser, J.W.**, Meisel, D.D. April 2018. Developing an automated bird song recognition system using wavelets and simple machine learning techniques. Oral presentation, Geneseo Recognizing Excellence, Achievement and Talent Day, Geneseo, NY.
- Doser, J.W.**, Hannam, K.M. October 2017. Using soundscape maps to assess a local soundscape. Poster presentation, Middle States Division American Association of Geographers Annual Meeting, Geneseo, NY.
- Doser, J.W.**, Hannam, K.M. October 2017. Analysis of seasonal temporal variation of soundscapes in western New York. Oral presentation, Northeast Regional Undergraduate Research, Scholarly and Creative Activity Conference for the Council of Public Liberal Arts Colleges, Mansfield, PA.
- Doser, J.W.**, Hannam, K.M. June 2017. Analysis of western New York forest soundscapes across spatial and temporal gradients. Twitter presentation, Biotweeps Twitter Conference.
- Doser, J.W.**, Meisel, D.D. April 2017. Quest for signs of life in the midst of chaos. Poster presentation, Geneseo Recognizing Excellence, Achievement and Talent Day, Geneseo, NY.
- Doser, J.W.**, Hannam, K.M. April 2017. Analysis of seasonal temporal variation of soundscapes in western New York. Oral presentation, Geneseo Recognizing Excellence, Achievement and Talent Day, Geneseo, NY.
- Doser, J.W.**, Hannam, K.M. April 2017. Analysis of seasonal temporal variation of soundscapes in western New York. Oral presentation, Northeast Natural History Conference, Cromwell, CT.
- Doser, J.W.**, Hannam, K.M. November 2016. Analyzing soundscape temporal variation in western New York as a potential assessment of biological diversity. Oral presentation, Rochester Academy of Science Fair, Rochester, NY.
- Doser, J.W.**, Hannam, K.M. August 2016. Use of soundscape recordings to analyze daily temporal variation in avian vocalizations in western New York. Poster presentation, North American Ornithological Conference, Washington D.C.

OUTREACH	Volunteer, Delta 4H Wildlife Club, East Lansing, MI	Spring 2019
	Guest Speaker, French Road Elementary School, Brighton, NY	2018
	• Discussed applications of Raspberry Pi computers and how to use them for sound recording.	
AFFILIATIONS	American Statistical Association	2019-current
	Ecological Society of America	2021-current
	American Ornithological Society	2021-current
	International Society of Ecoacoustics	2019-current

REFERENCES

Andrew O. Finley

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Elise F. Zipkin

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